

CANDIDATE'S NAME:.....

SIGNATURE

Index number							

545/1

Chemistry

Paper 1

2 HOURS



Apex Examination Council

MOCK EXAMINATIONS 2025

Uganda Certificate of Education

CHEMISTRY

Paper 1(Theory)

TIME: 2 Hours

INSTRUCTIONS TO CANDIDATES;

This paper consists of two sections section **A** and section **B**. It has six (6) items. Section A has two (2) **compulsory** items.

Section B has two parts, part I and part II, Attempt **ONLY** one item from each part.

Answers to section A must be written in the spaces provided while those to section B must be written on the answer booklet(s) provided. Any extra item will not be scored.

For official use only

ITEM	1	2	3	4	5	6
SCORE						

SECTION A

Answer all items in this section

ITEM 1

The under -17 National football team lost a game to Zambia in April, 2025 and the head coach attributes this loss to the severe muscle pains the boys had. Dr Ivan Kalika, the team doctor is planning to use muscle relaxants, Bacofen and clove oil from the Ugandan herbalist. He however warned the players about the risks of using the products for long.

Task

As a chemistry student:

a) Help the players to understand;

i) The category of the products the doctor suggested.

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ii) How the product will improve the players' fitness for the next game.

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b) Explain to the players the risks the team doctor meant.

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c) Evaluate the product to the players.

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ITEM 2

An alcoholic product, X_2H_5OH , containing 52.2% **X** is the chief ingredient in sanitizers due to its antiseptic properties. The public relations officer of one of the companies making products in Uganda has received many concerns from members of the public including among others, the composition of the ingredient.

Task

As a student of chemistry:

a) Explain to the public the category of the ingredient.

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b) Calculate the value of x in the ingredient. (H=1, O=16)

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- c) X completely burns in air providing a gaseous product XO_2 .

Explain the properties of the gaseous product that make it useful.

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- d) Highlight to the public the impacts of long-term use of the gaseous product.

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SECTION B

Part I

Answer one item from this part

Item 3

A group of senior 3 students from Ivy Progressive S.S went for a study trip to Muko hills in Kabale district, Western Uganda, Where large deposits of iron ores were discovered. The guide disclosed to the students that the government endorsed a Chinese company to extract iron. However, “Residents are worried about how the process is carried out and its environmental implications”. He said.

Task

Make a write up of the presentation you would make to the residents in a sensitization meeting with the Chinese company upon invitation.

Item 4

Health workers have raised concern about rising cases of hypertension (high blood pressure) in hospitals, which they say it is due to continued consumption of cooking oil. Farmers of palm trees in Kalangala are likely to make heavy losses.

Support material



Task

Advise the farmers on how they can benefit from their palm oil by making an alternative product.

Part II

Answer one item from this part

Item 5

An agricultural farm uses diesel-powered machines to till land and transport produces to the market.

Support material



The government officials have invited students' representatives from your school to farm to educate the workers about the fuel used.

Task

Write the information you would deliver to the workers as the key note speaker.

Item 6

Namanve forest reserve was cleared to set up an industrial park, harboring factories. This heavily altered the quality of air in the area due to waste fumes from factories and engines of heavy cargo trucks.

The District Environment Officer, Wakiso district has organized a meeting to create awareness to the industry operators

Support material



Task

As a learner of chemistry, write your message to the industry operators on behalf of the District Environment Officer.

END